INCM PROJECT PROPOSAL • QUESTION 27

How Does an Uncued, Inferred Prior Enter the Decision Process?

A Drift Diffusion Model Study of Mouse Perceptual Choice

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Evidence Accumulation

Decisions accumulate noisy sensory evidence over time until reaching a decision threshold (DDM).

Known Prior Routes

Explicit cued priors shift starting point (z); past choices shift drift rate (v).

The Unexplored Gap

IBL mice infer uncued probability blocks—entry route remains unmodeled in DDM frameworks.

RESEARCH SCOPE & PROBLEM FORMULATION

Dissecting the Inferred Prior Mechanism

Primary Research Questions

Part A (Baseline): Validate custom DDM simulator against mouse accuracy and RT across visual contrasts.

C1 (Main Novelty): Determine which mathematical route captures the inferred block prior (M1 vs M2 vs M3).

C2 (Boundary Shape): Test collapsing vs fixed boundaries on IBL wheel task with s_v, s_z variability controls.

M1: Starting Point Bias

Prior shifts starting position z. Predicts bias is strongest for fast choices and declines over time.

M2: Drift Rate Bias

Prior adds constant drift offset v_b. Predicts flat bias across the RT distribution.

M3: Dynamic Time-Growing Bias

Prior signal grows over elapsed decision time. Predicts strongest bias for slow choices.

COMPUTATIONAL METHODOLOGY & MATHEMATICAL SPECIFICATION

SDE Formulations, Mechanisms & Fitting Pipeline

1 Evidence Accumulation SDE

Stochastic Differential Equation:

$$x(t+\Delta t) = x(t) + v\Delta t + \sigma\sqrt{\Delta t}\xi(t)$$

where $\xi(t) \sim N(0,1)$ and $\sigma = 1$

Base Drift Rate Function:

$$v = k \cdot \text{sign}(c) \cdot |c|^{\beta} + v_h \cdot \text{prev_choice}$$

Signed contrast c, choice history v_h.

2 Prior Mechanisms (M1–M3) & Bounds

M1 Start: $x(0) = z \cdot \text{prior} \cdot a$

M2 Drift: $v \rightarrow v + v_b \cdot \text{prior}$

M3 Dynamic: $x(t) + \text{prior} \cdot b_{\max} \frac{t}{t+\tau_b}$

C2 Collapse: $\pm a \cdot \exp(-(t/\lambda)^{\gamma})$

C2 includes trial-to-trial variabilities s_v, s_z controls.

3 Simulation & Fitting Protocol

Integration: Euler–Maruyama ($\Delta t = 1\text{ms}$), cutoff at 5 s.

Quantile Fitting: Quantile Maximum Probability (QMP) matching observed vs simulated RT quantiles (0.1, 0.3, 0.5, 0.7, 0.9).

Optimization: Nelder–Mead optimization per mouse; model selection via BIC scoring.

Dataset: ~10 trained IBL mice (20/50/80 blocks) via ONE API.

LITERATURE & POSITIONING

Project Novelty & Key References

Closest Prior Work	What It Addressed	What It Left Open	Our Added Value
Fernandez et al. (2025)	DDM on history bias; previous choice → drift	Excluded probability blocks	**Fits DDMs directly to biased block sessions**
Hellmann & Forster (2026); Mulder et al.	Explicitly cued priors shift starting point	Priors that are inferred, not cued	**Tests uncued, experience-inferred prior**
Findling et al. (2023)	Mice infer block prior; neural representations	No decision-process (DDM) model	**Mechanism-level DDM account**
Hanks et al. (2011)	Time-growing prior bias (monkeys, humans)	Not tested in rodents	**Includes dynamic bias as competing mechanism**
Chakravarty et al. (2024)	Collapsing bounds in mouse nose-poke task	Wheel task untested	**Tests collapsing bounds on wheel task with s_v, s_z controls**

Data Repositories

International Brain Laboratory (2021) Mouse Visual Decision Dataset accessed via the public Python `one-api` package.

Key Citations

Fernandez et al. (2025) *bioRxiv* • Findling et al. (2023) *bioRxiv* • Hanks et al. (2011) *J. Neurosci* • Mulder et al. (2012) *J. Neurosci* • Urai et al. (2019) *eLife* • Chakravarty et al. (2024) *bioRxiv*

EXECUTION PLAN

4-Week Project Timeline & Workload Division

Week 1	**Person 1:** DDM simulator engine with M1–M3 & collapsing switches	**Person 2:** IBL data download, trial filtering & block labeling	**Validated DDM Simulator**
Week 2	**Person 1:** QMP fitting & parameter recovery pipeline	**Person 2:** Baseline fits & psychometric/chronometric curves	**Baseline Reproduction**
Week 3	**Person 1:** Fit M1–M3 prior models & bias functions (C1)	**Person 2:** Fit fixed vs collapsing boundaries with s_v, s_z (C2)	**Primary C1 & C2 Results**
Week 4	**Person 1:** Model sensitivity & robustness checks	**Person 2:** Final report writing & presentation deck assembly	**Final Submission**